

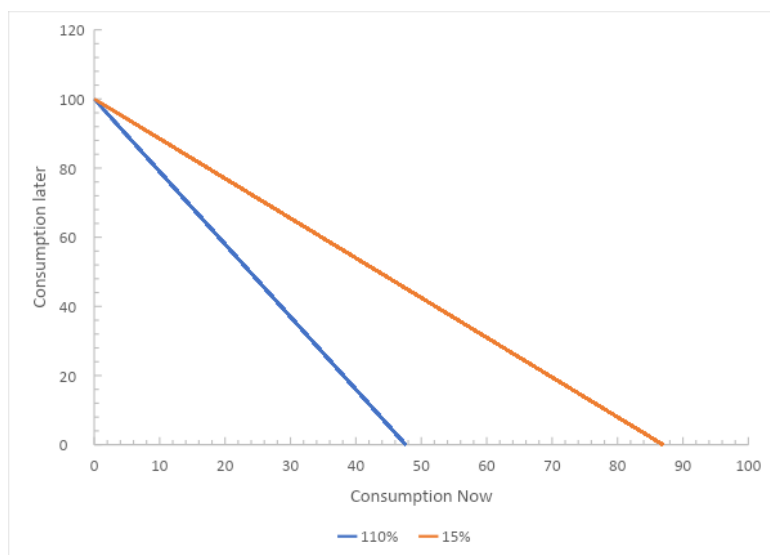
Principles of Economics

Problem Set 9

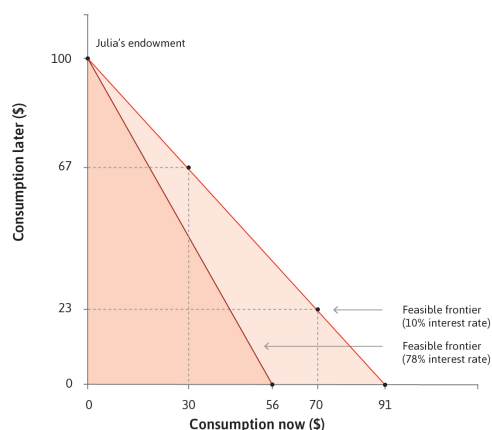
Universidad Carlos III de Madrid | Department of Economics

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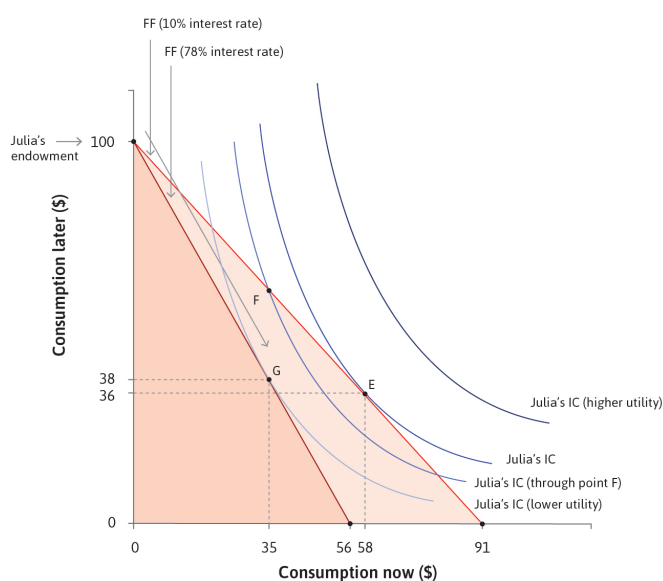
1. Construct Julia's feasible frontier by finding all the combinations of consumption now and next period, given her endowment and the interest rate.
 - a) Complete the table below, using the information provided. Round your answers to the nearest dollar.
 - b) Using your completed table, draw a diagram similar to Figure 9.3, showing the feasible frontier, the consumption amounts and the repayment amount.



2. Draw a set of indifference curves for Julia that reflect the following features:
 - a) She does not experience diminishing marginal returns to consumption but has intrinsic impatience. Would she then want to smooth her consumption?
 - b) She does not experience diminishing marginal returns to consumption, has no intrinsic impatience, and equally cares about consumption now and consumption later.
3. Use Figure 10.4 to show that the difference in current consumption at the lower and higher interest rate (at E and G), namely \$23, is composed of an income effect and a substitution effect. It will be helpful to review income and substitution effects from Unit 3 before doing this. Why do the income and substitution effects work in the same direction in this example?
4. The diagram depicts Julia's choice of consumption in periods 1 and 2. She has no income in period 1 and an income of \$100 in period 2. Based on this information, which of the following statements is correct?

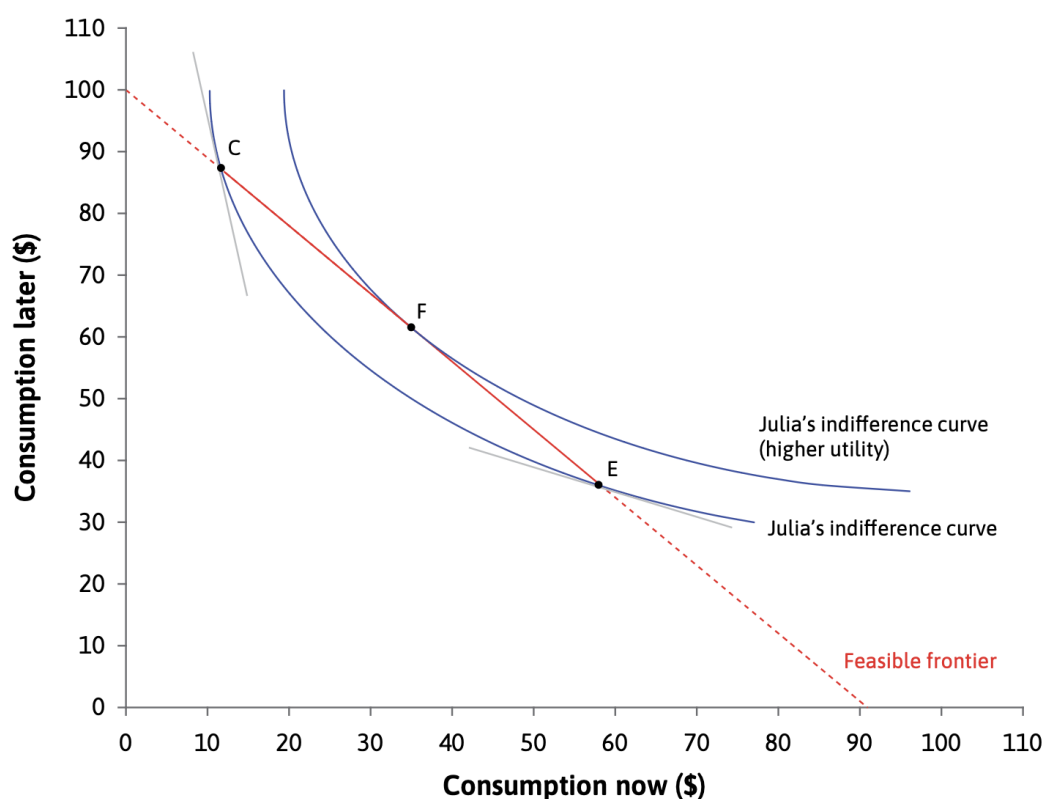


- a. When the interest rate is 10%, if Julia borrows \$23 in period 1 then she can consume \$75 in period 2.
 - b. When the interest rate is 10%, the marginal rate of transformation of goods from period 2 to period 1 is 0.1.
 - c. When the interest rate is 78%, the maximum that Julia can borrow in period 1 is \$56 (rounded to the nearest dollar).
 - d. When the interest rate is 78%, a consumption plan of \$30 in period 1 and \$67 in period 2 is in Julia's feasible set.
5. The diagram depicts Julia's indifference curve for consumption in periods 1 and 2. Based on this information, which of the following statements is correct?
- a. At consumption choice A, Julia values an extra unit of consumption in period 2 by 1.5 times the value of an extra unit of consumption in period 1.
 - b. Julia is impatient at all points on the indifference curve.
 - c. If Julia had less pure impatience, then her indifference curve would be steeper at A.
6. The diagram depicts Julia's choice of consumptions in periods 1 and 2. She has no income in period 1 and an income of \$100 in period 2. In scenario 1 the interest rate is 10%, while in scenario 2 it is 78%. Based on this information, which of the following statements is correct?



- a. Julia is able to consume more in period 2 at G under scenario 2 than at E under scenario 1, as her savings earn a higher interest in the former than in the latter.
 - b. Julia consumes less in period 1 at G under scenario 2 than at E under scenario 1, as she is less impatient at G.
 - c. The substitution and income effects of the interest rate rise partially offset each other, resulting in lower consumption in period 1 under scenario 2.
 - d. For the scenario of no income in period 1 and an income of \$100 in period 2, Julia is unambiguously worse off with an interest rate rise.
7. Which of the following statements about the banking system is correct?
- a. The policy rate is the rate at which the central bank lends in the money market.
 - b. The policy rate is determined by the supply and demand of the money market.
 - c. The bank lending rate is the rate at which the central bank lends to commercial banks.
 - d. The spread (markup) represents how the central bank's money is distributed amongst the commercial banks.
8. Figure 9.4 depicts an individual's indifference curves for consumption in periods 1 (now) and 2 (later). Based on this information, read the following statements and choose the correct option(s).

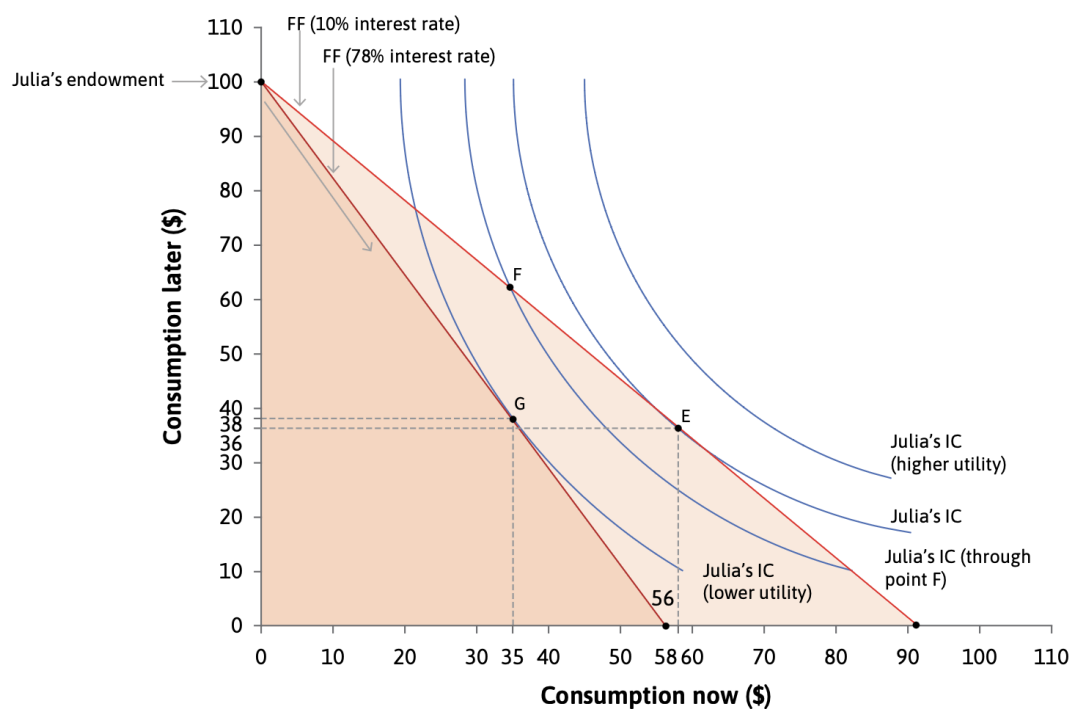
Figure 9.4



- a) The slope of the indifference curve is the marginal rate of substitution between the consumption in the two periods.
- b) The marginal return to consumption now is higher at E than at C.

- c) The individual's consumption is more equal across the two periods at E than at C. Therefore, the individual would prefer consumption choice E to C.
- d) Consuming exactly the same amount in the two periods is the individual's most preferred choice.
9. Figure 9.6 depicts Julia's choice of consumptions in periods 1 and 2. She has no income in period 1 (now) and an income of \$100 in period 2 (later). The current interest rate is 10%. Based on this information, read the following statements and choose the correct option(s).
- a) At F, the interest rate exceeds Julia's discount rate (degree of impatience).
- b) At E, Julia is on the highest possible indifference curve, given her feasible set.
- c) E is Julia's choice, as she is able to completely smooth out her consumption over the two periods and consume the same amount.
- d) G is not a feasible choice for Julia.

Figure 9.6

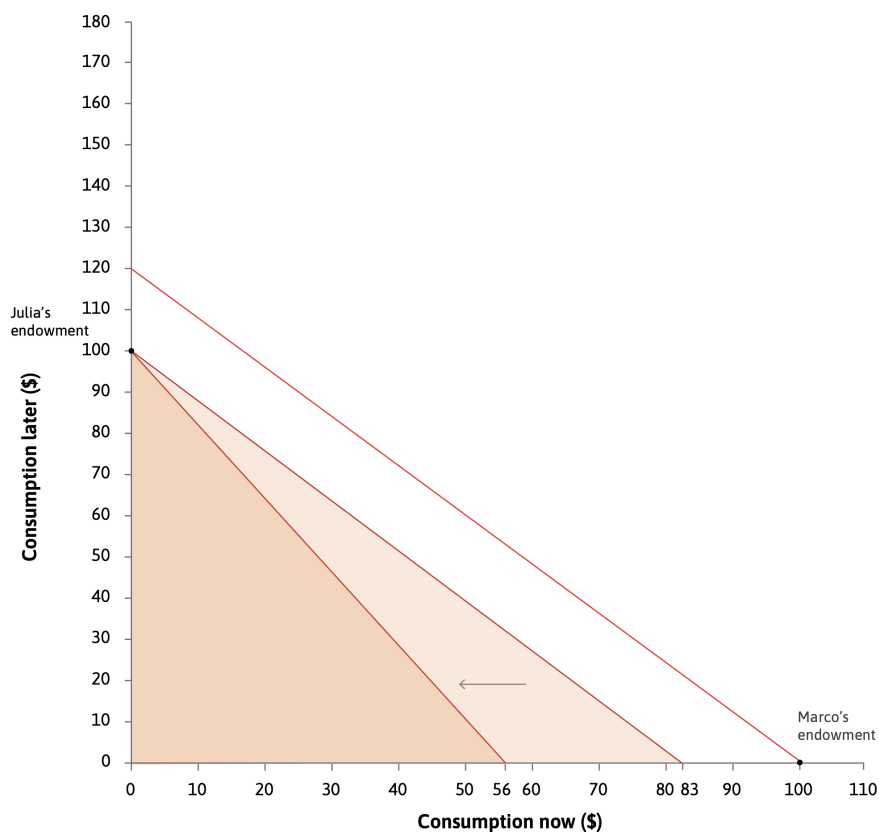


1. NUMERICAL PROBLEMS

10. Figure 9.14 shows the effect of an interest rate rise on Marco and Julia's feasible frontiers. Suppose that Marco and Julia's initial endowments are the same as in Figure 9.14, but the interest rate was initially 10% and increased to 45%. Based on this information, read the following statements and choose the correct option(s).
- a) With a 45% interest rate, Marco can consume \$80 now and \$29 later.
- b) With a 45% interest rate, Julia can consume \$58 now and \$18 later.
- c) If Marco were to save all of his initial endowment, he would have \$35 more to consume later under a 45% interest rate compared to a 10% interest rate.

- d) If Julia were to borrow and consume all of her endowment now, she would have \$20 less to consume now under a 45% interest rate compared to a 10% interest rate (rounded to the nearest whole number).

Figure 9.14



2. CONCEPTUAL QUESTIONS

11. Consider an individual's income over his or her lifetime from leaving school to retirement. Explain how an individual may move from a situation like Julia's to one like Marco's over the course of their lifetime (assume that their pure impatience remains unchanged over their lifetime).
12. Think about the income and substitution effects of a rise in the interest rate, as analysed in Exercise 10.2 and 10.3 in the book. Comment on whether a rise in the interest rate would be expected to reduce consumption expenditure in an economy in which a proportion of households are like Julia, and a proportion are like Marco.
13. Why do you think that banks tend to be more unpopular than other profit-making firms (Honda or Microsoft, for example)?
14. Many countries have policies that limit how much interest a moneylender can charge on a loan. What are likely to be the long-term effects of such laws?